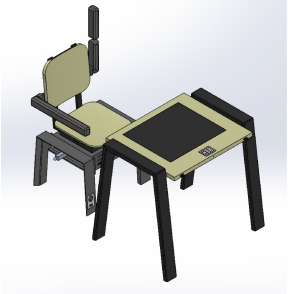
# POST TECH

Development of a modular workstation with integrated power

SolidWorks / QFD / Concept screening / Cost analysis | Academic Year 2025/2026



CAD representation of the concept: adjustable seat and independent desk.

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## EUR 119

estimated unit cost in the series production scenario

### OBJECTIVE

Develop an alternative to chairs with integrated writing tablets, focusing on ergonomics, modularity, electrical power and freedom of movement.

### MY CONTRIBUTION

I participated directly in every phase with the team: requirements, questionnaire, QFD, concepts, CAD and cost assessment. We developed the project jointly through brainstorming, without allocating exclusive tasks to individual members.

### ENGINEERING APPROACH

- Collection and classification of user needs; definition of metrics and target specifications using House of Quality.
- Concept generation and selection through screening and scoring; definition of architecture and interfaces.
- Assembly modelling in SolidWorks and estimation of component, manufacturing and assembly costs.

### RESULTS

The selected concept includes an independent desk, an adjustable seat, inter-unit connections and modular accessories. The initial total estimate is EUR 170 per unit; the series production scenario assumes a 30% reduction to EUR 119.

### SCOPE AND LIMITATIONS

Concept development and preliminary assessment: costs are project estimates, rather than results from established industrial production.

### PROJECT TEAM

Francesco Paolucci / Davide Calvelli / Edoardo Nicasi Zanetti / Tommaso Pierini